








Advanced Time Series Analysis and Forecasting

Chapter 14: Causal inference for time series



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Acronyms used in this material (1/2)

- **ADH**: Abadie, Diamond and Hainmueller
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CAR**: Cumulative Abnormal Return
- **CCM**: Convergent Cross Mapping
- **CS**: Callaway and Sant'Anna (estimator)
- **DML**: Double/Debiased Machine Learning
- **ECOFIN**: Economic and Financial Affairs Council (of the EU)
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **EU-26**: the 26 EU countries other than Romania
- **GDP**: Gross Domestic Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **ITS**: Interrupted Time Series
- **LLM**: Large Language Model
- **LP**: Local Projection

Acronyms used in this material (2/2)

- **MCI**: Momentary Conditional Independence
- **MCMC**: Markov Chain Monte Carlo
- **MSPE**: Mean Squared Prediction Error
- **OECD**: Organisation for Economic Co-operation and Development
- **OLS**: Ordinary Least Squares
- **PC1**: first Principal Component
- **PCMCI**: PC algorithm + Momentary Conditional Independence
- **RMSPE**: Root Mean Squared Prediction Error
- **SC**: Synthetic Control
- **SDID**: Synthetic Difference in Differences
- **SEC**: Securities and Exchange Commission
- **SVAR**: Structural Vector AutoRegression
- **TE**: Transfer Entropy
- **TSA**: Time Series Analysis
- **TWFE**: Two-Way Fixed Effects
- **UK**: United Kingdom
- **US**: United States
- **USA**: United States of America
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax
- **VIX**: CBOE Volatility IndeX

Today's question and route

- **Question**: when does a time-series analysis tell us what *would have happened* without a shock or a policy, and how sure can we be?
  - prediction is not intervention: a variable can forecast another without causing it
- **Route** of the chapter
  - Granger causality, potential outcomes for time series, transfer entropy
  - causal discovery: PCMCI and convergent cross mapping, with their limits
  - interrupted time series and event studies with dependent errors
  - synthetic control and its successors; staggered DiD; CausalImpact; DML
- Prerequisites: Seminar 14 and the slide Known from TSA and Chapter 3, and new here

Learning outcomes

- Distinguish Granger causality from a dynamic causal effect, and state the conditions under which a time-series estimand has a causal meaning
- Run conditional Granger tests with HAC covariance, estimate transfer entropy, and read the output of PCMCI and convergent cross mapping critically
- Estimate interrupted-time-series and event-study effects with standard errors that respect serial dependence
- Build synthetic controls (classic, demeaned, augmented, synthetic DiD), run placebo inference, and replicate a published result
- Use CausalImpact-type state space counterfactuals and judge the identifying assumptions of every method honestly

Reading, data and tools

- Causality and time series: Granger (1969); Sims (1972); Rambachan and Shephard (2021); Bojinov and Shephard (2019); Angrist, Jordà and Kuersteiner (2018); Stock and Watson (2018)
  - discovery: Schreiber (2000); Runge et al. (2019a); Runge et al. (2019b); Sugihara et al. (2012)
- Policy evaluation
  - synthetic control: Abadie, Diamond and Hainmueller (2010); Abadie, Diamond and Hainmueller (2015); Abadie (2021); Ben-Michael, Feller and Rothstein (2021); Arkhangelsky et al. (2021)
  - staggered DiD and CausalImpact: Callaway and Sant'Anna (2021); Brodersen et al. (2015); survey Athey and Imbens (2017)
- Python Quantlets of this chapter: Quantlets/Ch_14
  - `numpy`, `scipy` (synthetic control weights by quadratic programming) and `statsmodels` (state space); every estimator written out, no black box
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source, sample	Use
HICP inflation, 27 EU countries; HICP at constant tax rates	Eurostat (2026a), Eurostat (2026b): monthly, 2015 – August 2026	ITS, synthetic control for Romania
GDP per capita, West Germany and 16 OECD countries	Abadie, Diamond and Hainmueller (2014): 1960–2003	replication of ADH (2015)
Real GDP, UK and 23 OECD countries	OECD (2026): quarterly, 1995–2019	Brexit doppelganger
Daily index and crypto prices; EUR/RON	EODHD; BNR reference rate; to 18 September 2026	Granger, PCMCI, event study, CausalImpact

Two languages of causality



Judea Pearl, Turing Award 2011

Photo: Better Than Bacon (2013); CC BY 2.0; Wikimedia Commons

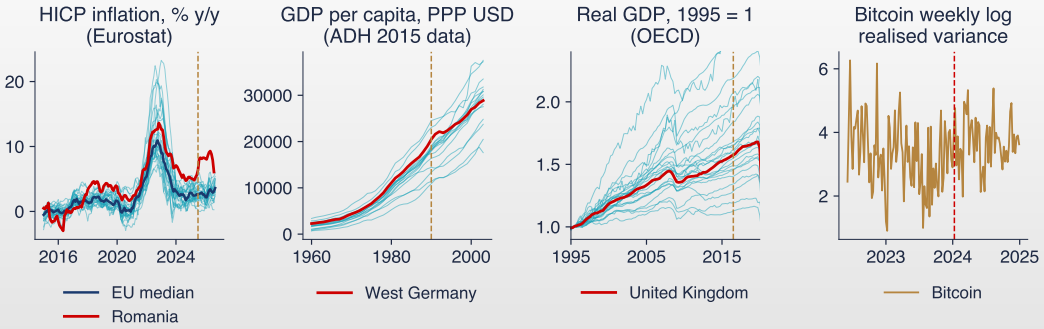


Guido Imbens, Nobel Prize in Economics 2021

Photo: Filetime (2022); CC0; Wikimedia Commons

- **Structural causal models** Pearl (2009)
  - graphs and the $do(\cdot)$ operator: $do(X = x)$ sets X by intervention instead of observing it
  - conditions under which a causal effect is identified from observational data
- **Potential outcomes** Rubin (1974), Imbens and Rubin (2015)
  - $Y(1)$, $Y(0)$: the outcomes of one unit with and without treatment; only one of them is observed
  - assignment mechanisms; design before analysis
- **Time series** add order, dependence and a single realisation
  - one country, one history, one treatment path
  - Granger and PCMCI speak the graph language; synthetic control, DiD and CausalImpact speak potential outcomes

Four case studies



- Romanian inflation against 26 EU countries; West Germany against the OECD sample; UK real GDP against 23 OECD countries; Bitcoin realised variance around the spot ETF approval

ATS_ch14_romania

Interpreting the four case studies

- Romania: annual HICP inflation 5.6% in June 2025, 8.2% in August 2025, peak 9.3% in May 2026; 6.1% in August 2026 (EU median 3.1%)
- Each case has one treated unit and a pool of untreated units or controls observed over a long pre-period: the comparative-case-study design
- Romania is outside the range of the donors before the treatment (the highest inflation in the EU): a warning for every method that interpolates
- Bitcoin volatility is noisy and the controls are weak: the setting where an honest analysis may find nothing

Known from TSA and Chapter 3, and new here

- Known: the bivariate Granger test in a VAR (TSA, Chapter 6); SVAR identification, local projections and proxy SVAR (Chapter 3); HAC (Chapter 0); Kalman filter and BSTS (Chapter 6)
- New: what these tools can and cannot say about interventions
  - potential outcomes for time series; non-anticipation; when an impulse response is a causal effect
  - nonlinear and graph-based discovery; comparative case studies with one treated unit
- Replications: Abadie, Diamond and Hainmueller (2015, Table 1, Figures 2–5); Born et al. (2019, Table 2, Figure 2); Sugihara et al. (2012, Figure 3)

Granger causality: the definition (1/2)

- Granger (1969): x **Granger-causes** y if the past of x lowers the mean squared error of the best one-step forecast of y

$$\mathbb{E}\left[(y_{t+1} - \mathbb{E}[y_{t+1} \mid \mathcal{I}_t])^2\right] < \mathbb{E}\left[(y_{t+1} - \mathbb{E}[y_{t+1} \mid \mathcal{I}_t \setminus x])^2\right]$$

  - $\mathcal{I}_t$: all the information available up to t; $\mathcal{I}_t \setminus x$: the same information without the past of x
  - $\mathbb{E}[y_{t+1} \mid \cdot]$: the conditional mean, i.e. the best forecast in mean squared error

- In practice $\mathcal{I}_t$ is a finite set of lags in a regression

$$y_t = c + \sum_{j=1}^{p} a_j y_{t-j} + \sum_{j=1}^{p} b_j x_{t-j} + \sum_{j=1}^{p} \gamma_j' z_{t-j} + u_t$$

  - p: the number of lags; a_j, b_j: coefficients on the lags of y and of x; z_t: a vector of other conditioning series, with coefficient vectors γ_j; u_t: the error
  - null hypothesis of Granger non-causality: H_0: $b_1 = \cdots = b_p = 0$

Granger causality: the definition (2/2)

- **Wald statistic**: a quadratic form of the estimated lag coefficients of x

$$W = \hat{b}' \hat{V}_b^{-1} \hat{b} \;\rightarrow\; \chi^2_p \quad \text{under } H_0$$

  - $\hat{b} = (\hat{b}_1, \ldots, \hat{b}_p)'$; $\hat{V}_b$: its estimated covariance matrix; reject H_0 for a large W (small p-value)
  - $\hat{V}_b$ HAC (Chapter 0) when u_t is heteroskedastic, as for returns Newey and West (1987)
- Integrated or cointegrated series Toda and Yamamoto (1995)
  - add as many extra lags as the maximal order of integration and test only the first p lags
- **Strength**: the Geweke measure Geweke (1982), decomposable by frequency Breitung and Candelon (2006) (Chapter 11)

$$F_{x \to y} = \ln\left(\sigma^2_{\text{restricted}} / \sigma^2_{\text{full}}\right)$$

  - $\sigma^2_{\text{restricted}}$, σ^2_{full}: residual variances of the regression without and with the lags of x; $F_{x \to y} \geq 0$, and 0 means no Granger causality

Granger causality is about prediction

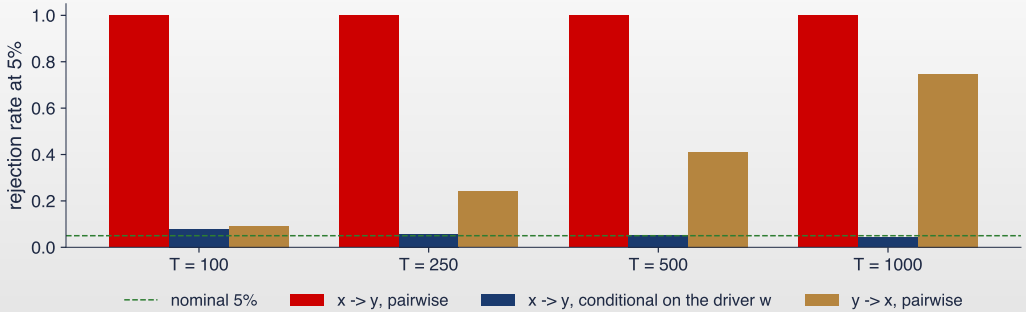


Clive Granger, Nobel Prize in Economics 2003

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

- Four ways in which x Granger-causes y without causing it
  - **common driver** z, omitted, reaching x and y at different lags
  - **expectations**: asset prices move before the events they anticipate (stock prices "cause" GDP; Sims (1972))
  - **timing**: different closing hours, aggregation and sampling turn instantaneous links into lagged ones
  - **measurement error** in y that x helps to filter
- And the reverse: a true effect can be invisible to the test
  - nonlinear, contemporaneous, or offset by policy feedback

A common driver creates Granger causality



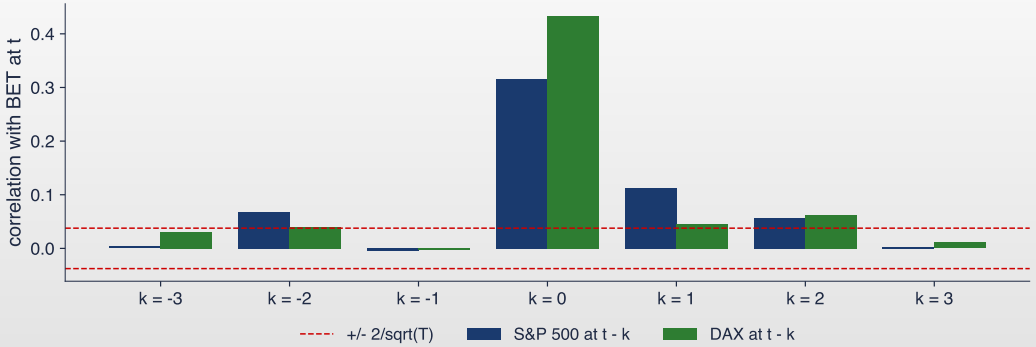
- Simulated system: a driver $w_t = 0.9 w_{t-1} + \eta_t$ (AR(1), $\phi = 0.9$); $x_t = w_{t-1} + e_t$, $y_t = w_{t-3} + u_t$; η_t, e_t, u_t: independent noise
- x has no effect on y; tests with $p = 2$ lags (pairwise) and $p = 4$ (conditional on w), 500 replications

ATS_ch14_granger

Interpreting the common-driver experiment

- Pairwise, x "Granger-causes" y in 100% of the samples already at $T = 100$: x carries news about w two periods before y does
- Conditional on the driver the rejection rate falls to the nominal level (4% at $T = 1000$): Granger causality is relative to the information set
- The reverse test also rejects more and more (9% at $T = 100$, 75% at $T = 1000$): the past of y is informative about the persistent driver as well
- Larger samples make a spurious finding more certain, not less: power is not validity

Lead and lag between New York, Frankfurt and Bucharest



- Daily log returns, common trading days 6 January 2015 – 18 September 2026 ($T = 2\,815$); correlation of the BET at t with the S&P 500 and the DAX at $t - k$

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Granger tests on the three indices

Hypothesis ($p = 2$)	**Wald (HAC)**	***p*-value**	**conditioning**
S&P 500 $\not\to$ BET	15.3	<0.001	–
BET $\not\to$ S&P 500	4.5	0.105	–
S&P 500 $\not\to$ BET	9.9	0.007	DAX
DAX $\not\to$ BET	1.6	0.449	–
S&P 500 $\not\to$ DAX	18.3	<0.001	–

- The classical (non-HAC) Wald for S&P 500 $\to$ BET is 34.3: ignoring heteroskedasticity would artificially double the evidence
- Interpretation: New York closes after Bucharest and Frankfurt; the lag-1 correlation (0.11) is news that arrives after the Bucharest close, not an effect of US prices on Romanian firms
- The DAX, which closes with Bucharest, shows contemporaneous correlation (0.43) and no lagged effect: timing creates the Granger result

Potential outcomes for time series (1/2)

- **Assignment path**: the sequence of treatments (or shocks) $w_{1:T} = (w_1, \ldots, w_T)$ Rambachan and Shephard (2021), Bojinov and Shephard (2019)
  - w_t: the treatment at t, binary (policy on or off) or continuous (the size of a shock)
- **Potential outcome** $Y_t(w_{1:T})$: the value Y_t would take if the treatment path were $w_{1:T}$
  - only the potential outcome of the path that actually happened is observed
- **Non-anticipation**: the outcome today depends only on treatments up to today

$$Y_t(w_{1:T}) = Y_t(w_{1:t})$$

  - announcements violate it, unless the announcement itself is defined as the treatment

Potential outcomes for time series (2/2)

- **Dynamic causal effect** at horizon h: the change in Y_{t+h} when only the treatment at t changes from w' to w

$$\tau_{t,h} = Y_{t+h}(w_{1:t-1}, w, w_{t+1:t+h}) - Y_{t+h}(w_{1:t-1}, w', w_{t+1:t+h})$$

  - the past treatments $w_{1:t-1}$ and the later ones $w_{t+1:t+h}$ are kept fixed
  - one history is observed: only averages over time (or over units) of $\tau_{t,h}$ are estimable
- Rambachan and Shephard (2021): suppose w_t is **sequentially as good as random**, i.e. independent of the potential outcomes given the past
  - then the impulse response and the local projection coefficient are weighted averages of $\tau_{t,h}$, without assuming linearity
- The hard part is never the regression; it is the claim that the shock is unpredictable from the potential outcomes

Identifying dynamic causal effects in macroeconomics

- **Narrative and high-frequency shocks** as instruments: local projections Jordà (2005) and proxy SVAR Stock and Watson (2018) (Chapter 3)
  - the instrument must be relevant, contemporaneously exogenous and uncorrelated with leads and lags of the other shocks Stock and Watson (2018)
- **Propensity scores for policy**: Angrist, Jordà and Kuersteiner (2018) model the Fed's decision given the forecasts it saw, and reweight outcomes (inverse probability weighting)
  - identification by selection on observables: the Fed's information set is observed through the Greenbook
- **Time-series experiments**: Bojinov and Shephard (2019) give exact randomisation tests for one unit treated at random times (trading experiments)
- Local projections and VARs estimate the same population responses Plagborg-Møller and Wolf (2021): the choice is about bias and variance, not identification

Transfer entropy (1/2)

- Schreiber (2000): the information that the past of x adds about y_t, beyond the past of y

$$TE_{x\to y} = I\left(y_t;\, x_{t-1}^{(l)} \mid y_{t-1}^{(k)}\right) = \mathbb{E}\left[\ln \frac{p(y_t \mid y_{t-1}^{(k)}, x_{t-1}^{(l)})}{p(y_t \mid y_{t-1}^{(k)})}\right] \geq 0$$

- Notation
  - $y_{t-1}^{(k)} = (y_{t-1}, \ldots, y_{t-k})$: the last k values of y; $x_{t-1}^{(l)}$: the last l values of x
  - $p(\cdot \mid \cdot)$: a conditional density; $I(A; B \mid C)$: the conditional mutual information of A and B given C, in nats (natural logarithm)
- $TE_{x\to y} = 0$ if and only if x adds nothing to the predictive distribution of y
  - Granger non-causality in distribution, not only in the mean

Transfer entropy (2/2)

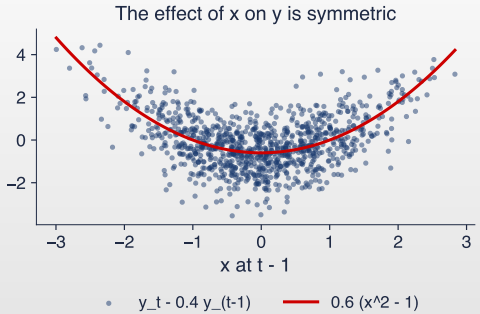
- **Gaussian case** Barnett, Barrett and Seth (2009): transfer entropy is half the Geweke measure, so the linear Granger test is a TE test

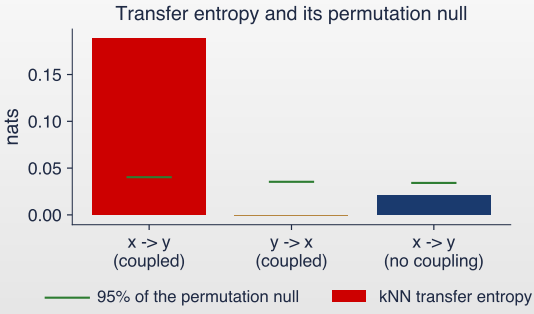
$$TE_{x\to y} = \frac{1}{2}\ln\left(\sigma^2_{\text{restricted}}/\sigma^2_{\text{full}}\right) = \frac{1}{2}F_{x\to y}$$

  - proof in the Appendix

- **Nonparametric estimation**: k-nearest-neighbour estimators Kraskov, Stögbauer and Grassberger (2004), Frenzel and Pompe (2007)
  - densities are replaced by distances to the nearest neighbours in the joint space
  - significance by permuting the source in blocks, which keeps its autocorrelation
- The same caveats as for Granger: an information flow is not an intervention

A nonlinear coupling that linear Granger misses





- Simulated: $y_t = 0.4 y_{t-1} + 0.6(x_{t-1}^2 - 1) + u_t$, x a Gaussian AR(1) with unit variance, u_t standard Normal noise, $T = 1\,000$
- $x_{t-1}^2 - 1$ has mean zero and no correlation with x_{t-1}; kNN TE with $k = 5$ neighbours, 199 block permutations

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Interpreting transfer entropy

- Linear Granger: p-value 0.86; Gaussian TE 0.05×10^{-3} nats: the symmetric effect averages out in a linear regression
- kNN TE $x \to y$: 0.189 nats, permutation p 0.005; reverse direction -0.001 (p 0.76); without coupling 0.021 (p 0.26)
- On the markets of the previous slide the Gaussian TE from the S&P 500 to the BET is 6.1×10^{-3} nats: significant, but tiny as information
- TE needs much more data than Granger and a careful choice of k, history lengths and permutation blocks; pre-register them

Recap: Granger causality and causal effects

- Granger causality is incremental predictability relative to an information set; enlarging the set can create or destroy it
- A causal effect needs potential outcomes, non-anticipation and an assignment that is as good as random given the past
- Transfer entropy extends Granger to distributions; for Gaussian processes the two coincide

Time series graphs (1/2)

- **Nodes** X_t^i: variable i at time t; an **arrow** $X_{t-\tau}^i \to X_t^j$ when $X_{t-\tau}^i$ is a direct cause of X_t^j given the rest of the past Runge et al. (2019b)
  - $\tau \geq 1$: the lag of the link; $\tau_{\max}$: the largest lag considered
  - the graph is the same at every t (causal stationarity); arrows only go forward in time
- Three assumptions turn independences into arrows Spirtes, Glymour and Scheines (2000)
  - **causal Markov**: each variable is independent of its non-effects given its parents (direct causes)
  - **faithfulness**: no independence arises by cancellation of effects
  - **causal sufficiency**: no hidden common causes

Time series graphs (2/2)

- Under these assumptions an arrow is a conditional dependence given the whole past

$$X^i_{t-\tau} \to X^j_t \iff X^i_{t-\tau} \not\perp X^j_t \mid \mathbf{X}^-_t \setminus \{X^i_{t-\tau}\}$$

  - $\perp$: independence ($\not\perp$: dependence); $\mathbf{X}^-_t$: all variables at lags $1, \dots, \tau_{\max}$ before t
  - this is the full conditional (VAR) Granger test
- With N variables and $\tau_{\max}$ lags the conditioning set has $N\tau_{\max}$ elements
  - the power of the test collapses in high dimension: the motivation for PCMCI

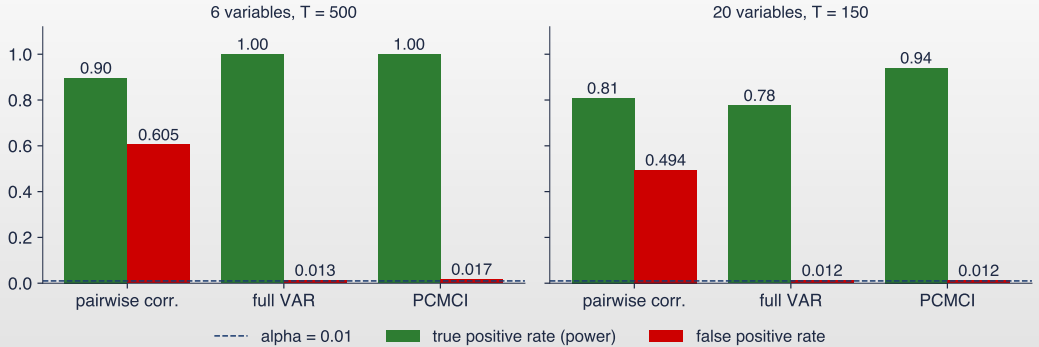
PCMCI

- Runge et al. (2019a): two steps, each a sequence of conditional independence tests (partial correlation here; nonlinear tests possible)
- **PC1** (condition selection), for each X_t^j
  - start from all lagged candidates; remove those independent of X_t^j given the p strongest others, $p = 1, 2, \ldots$
  - the result $\hat{\mathcal{P}}(X_t^j)$ is a superset of the parents of X_t^j
- **MCI** (momentary conditional independence): test each link given the parents of both ends

$$X_{t-\tau}^i \perp X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\},\ \hat{\mathcal{P}}(X_{t-\tau}^i)$$

  - conditioning on the parents of the **source** $X_{t-\tau}^i$ removes its autocorrelation: false positives stay near α
  - conditioning sets stay small, so power is kept in high dimension
- Hidden confounders, contemporaneous links, nonstationarity and measurement error break the guarantees; the output is a hypothesis about a graph

PCMCI against correlation and the full VAR



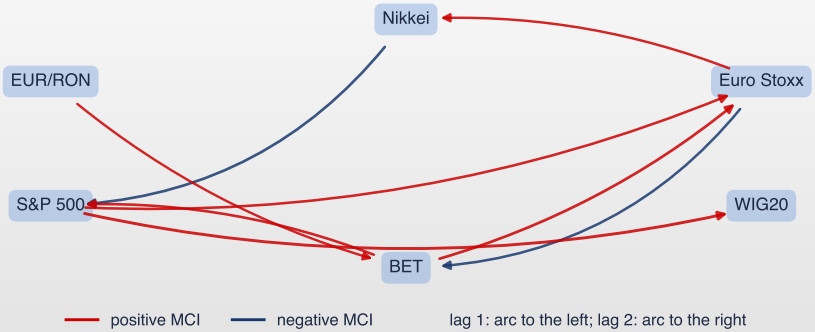
- A known lagged system (a persistent common driver, chains and a collider), $\tau_{\max} = 3$, $\alpha = 0.01$, 50 replications; right: 14 extra independent AR(1) series

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Interpreting the discovery experiment

- Lagged correlations find most true links but flag 0.605 of the absent ones: autocorrelation and the common driver make everything correlated
- Six variables, $T = 500$: the full VAR and PCMCI are equivalent (power 1.00 and 1.00; false positives 0.013 and 0.017)
- Twenty variables, $T = 150$: the VAR conditions on 60 regressors and its power falls to 0.78; PCMCI keeps 0.94 with false positives 0.012
- The advantage of PCMCI is a high-dimensional one; in small systems a well-specified VAR is enough

A discovered graph of market volatility



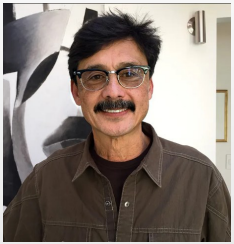
- PCMCI on weekly log realised variances (sum of squared daily returns), 13 January 2008 – 20 September 2026, $T = 973$ weeks, $\tau_{\max} = 2$, $\alpha = 0.01$; weeks remove the ordering of daily closes

ATS_ch14_discovery

Interpreting the volatility graph

- Lagged correlation tests are significant for 56 of the 60 possible lagged cross links; PCMCI keeps 8
- Volatility is a common factor: most of the raw dependence is explained by each market's own past and by the shared global shock within the week
- The surviving arrows are small partial correlations; contemporaneous spillovers within the week are invisible to a lagged-only analysis
- A hidden global factor (news, VIX) violates causal sufficiency: read the graph as a map of predictive links, not of spillover mechanisms

Convergent cross mapping



George Sugihara

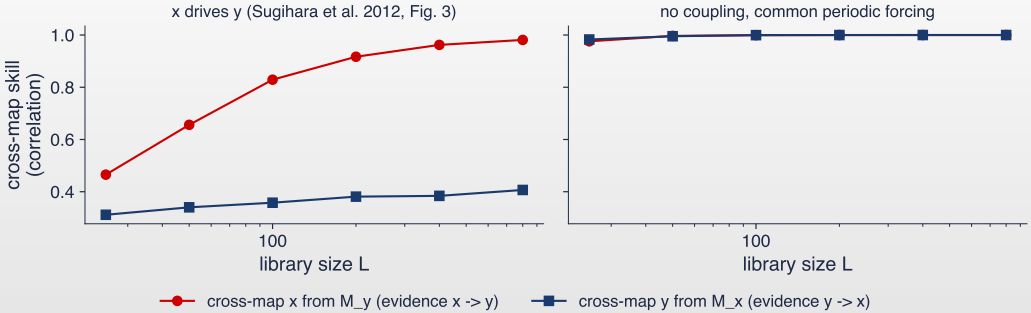
Photo: J Park (2015); CC BY-SA 4.0; Wikimedia Commons

- Deterministic coupled dynamics: Granger's separability fails, because the information of the cause is already in the effect Sugihara et al. (2012)
- Takens Takens (1981): the delay vectors of y reconstruct the attractor of the whole system

$$M_y = \{(y_t, y_{t-\tau}, \ldots, y_{t-(E-\mathbf{1})\tau})\}$$

  - E: the embedding dimension; τ: the delay between coordinates
- If x drives y, the neighbours of a point on M_y identify x_t: **cross-map** x from M_y
  - simplex projection: a weighted average of x at the $E+1$ nearest neighbours; skill: the correlation ρ between estimate and truth
- Criterion: the skill must **converge** (increase) with the library size L, the number of points used to build M_y
- Note the direction: skill of $M_y \to x$ is evidence for $x \to y$

Cross mapping: a coupling and a common forcing



- Coupled logistic maps of Sugihara et al. (2012): $x_{t+1} = x_t(r_x - r_x x_t - \beta_{xy} y_t)$, $y_{t+1} = y_t(r_y - r_y y_t - \beta_{yx} x_t)$
- $r_x = 3.8$, $r_y = 3.5$: growth rates (chaotic regime); $\beta_{yx} = 0.32$: effect of x on y; $\beta_{xy} = 0$: no effect of y on x; right: two uncoupled maps with the same periodic forcing; $E = 2$

ATS_ch14_discovery

Interpreting cross mapping

- Left: skill for $x \to y$ rises from 0.47 ($L = 25$) to 0.98 ($L = 800$); the reverse stays near 0.41: the published pattern
- Right: no coupling at all, yet both directions reach 0.999 and 0.999: a shared periodic driver synchronises the maps and CCM reports bidirectional causality
- Known critiques: seasonality and synchrony Baskerville and Cobey (2017), noise and external forcing Mønster et al. (2017), the choice of lag Ye et al. (2015)
- Use CCM for low-noise nonlinear systems, with surrogate tests that preserve seasonality; never as a black-box causality detector for economic series

Recap: Causal discovery

- A discovered graph is only as good as causal Markov, faithfulness and sufficiency; hidden common drivers are the rule in economics
- PCMCI controls false positives under autocorrelation and keeps power in high dimension
- CCM targets deterministic coupling; synchrony and common forcing produce false bidirectional links

Interrupted time series (1/2)

- One series, an intervention at T_0 Lopez Bernal, Cummins and Gasparrini (2017); step 1: model the pre-period

$$y_t = f(t, \text{season}, x_t; \theta) + u_t, \qquad t \leq T_0$$

  - f: trend, seasonal effects and covariates x_t, with parameters θ; u_t: the error
- Step 2: the effect is the gap between the outcome and the extrapolated model

$$\hat{\tau}_t = y_t - f(t, \cdot; \hat{\theta}), \qquad t > T_0$$

  - segmented regression is the special case with level and slope dummies after T_0
- **Identification**
  - nothing else changes at T_0; the pre-period model would have continued; no anticipation

Interrupted time series (2/2)

- **Inference with dependent errors**: the variance of the cumulative effect over n periods

$$\mathrm{Var}\Big(\sum_{t=T_0+1}^{T_0+n} \hat{\tau}_t\Big) \approx n\,\Omega + a'\hat{V}_\theta a$$

  - Ω: the long-run variance of u_t (Chapter 0); $\hat{V}_\theta$: the covariance of $\hat{\theta}$; a: the sum of the regressors of f over the n periods
  - first term: noise of the outcomes; second term: estimation error of the counterfactual
- With AR(1) errors, autocorrelation ρ and $\sigma^2 = \mathrm{Var}(u_t)$

$$\Omega = \sigma^2 \frac{1+\rho}{1-\rho}$$

  - i.i.d. standard errors are too small by the factor $\sqrt{(1+\rho)/(1-\rho)}$, e.g. 1.7 for $\rho = 0.5$

Romania 2025: two measures one month apart

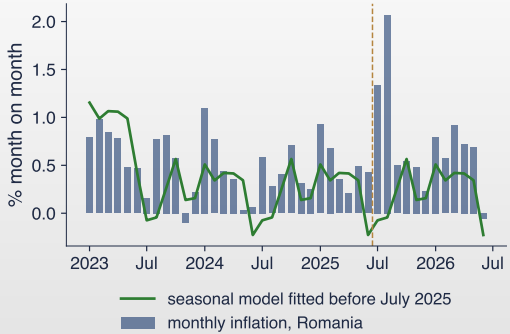


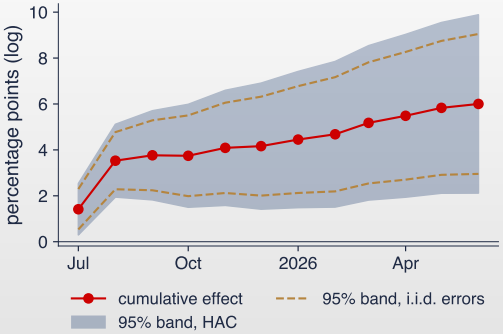
Palace of the Parliament, Bucharest

Photo: William John Gauthier (2017); CC BY-SA 2.0; Wikimedia Commons

- **1 July 2025**
  - the cap on household electricity prices (in force since 2022) ends
  - the household gas cap continues
- **7 July 2025**
  - the government assumes responsibility for a fiscal package in Parliament (Law 141/2025, published 25 July)
- **1 August 2025**
  - standard VAT 19% $\rightarrow$ 21%; reduced rates 5% and 9% $\rightarrow$ 11%; excise duties up
- Timing alone cannot separate the two; the HICP at constant tax rates can isolate the tax part
- Question: how much did the two measures add to Romanian inflation over the following year?

An interrupted time series for Romanian inflation





- Monthly HICP inflation (log change); pre-period model: 12 month effects and a step for July 2021 – June 2023, fitted on 126 months to June 2025; effects July 2025 – June 2026

ATS_ch14_its_event

Interpreting the interrupted time series

- July 2025: +1.41 pp above the seasonal norm; August 2025: +2.11 pp; later months +0.25 pp on average
- Cumulative effect over twelve months 6.0 pp; standard error 1.55 with i.i.d. errors, 1.98 with HAC (residual autocorrelation 0.33, long-run variance 1.63 times σ^2)
- The ITS counterfactual is Romania's own seasonal norm: it ignores everything else that happened in Europe after June 2025
- A control group is needed to remove common shocks: the synthetic control of Section 5 uses 26 EU countries

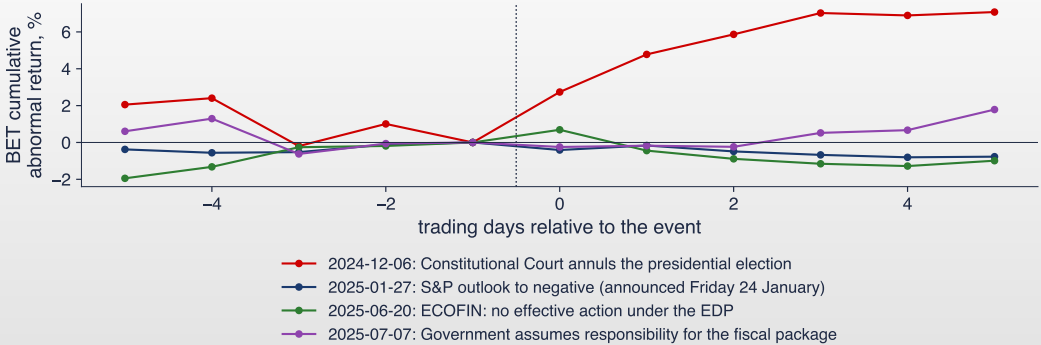
Event studies with dependent errors

- MacKinlay (1997), the **market model** estimated on days $[-250, -11]$ before the event

$$r_t = \alpha + \beta r_t^m + \varepsilon_t, \qquad AR_t = r_t - \hat{\alpha} - \hat{\beta} r_t^m, \qquad CAR[0,k] = \sum_{t=0}^{k} AR_t$$

  - r_t: the return of the asset; r_t^m: the market return; AR_t: the abnormal return; $CAR[0,k]$: the cumulative abnormal return over days 0 to k
- Variance of the CAR
  - $(k+1)\sigma^2$ under i.i.d. errors; $(k+1)\Omega$ with serial correlation (Ω: long-run variance)
  - event-induced variance and clustered events need cross-sectional or bootstrap corrections
- Identification: the event is a surprise on day 0 and nothing else happened
  - anticipated news is priced before the window
- Here: the BET index against the Euro Stoxx 50, four Romanian political and fiscal events of 2024–2025, window $[0, 2]$

Romanian stocks around political and fiscal news



⊡ Cumulative abnormal return of the BET, normalised at day −1; market model on the Euro Stoxx 50; rating actions announced after the close on a Friday are dated on the next Monday

ATS_ch14_its_event

Interpreting the event study

- Annulment of the presidential election (6 December 2024): CAR[0, 2] +5.9%, $t = 5.25$ (i.i.d.) and 5.08 (HAC): the market priced a lower political risk
- S&P negative outlook: -0.5% ($t = -0.33$); ECOFIN decision: -0.9% ($t = -0.52$); fiscal package: -0.2% ($t = -0.14$)
- Fiscal news had been discussed for months: no surprise on day 0, no abnormal return; a null result here says nothing about the economic effect of the measures
- HAC and i.i.d. standard errors are close for daily returns (little autocorrelation); they differ for monthly macro data (previous section)

Recap: Interrupted time series and event studies

- ITS and event studies compare a series with its own extrapolated past: valid only if nothing else changed at the same time
- Cumulative effects need the long-run variance of the errors, not σ^2
- Anticipation moves the effect before the window: dating the news is part of the identification

The synthetic control estimator (1/2)

- Units: $j = 1$ treated, $j = 2, \ldots, J+1$ the **donor pool**; treatment after period T_0 Abadie and Gardeazabal (2003), Abadie, Diamond and Hainmueller (2010)
  - $Y_{jt}(1)$, $Y_{jt}(0)$: potential outcomes of unit j at t with and without the treatment
- Target: the effect on the treated unit after T_0

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \qquad t > T_0$$

  - $Y_{1t}(1) = Y_{1t}$ is observed; $Y_{1t}(0)$ must be estimated
- **Synthetic control**: a weighted average of the donors stands in for $Y_{1t}(0)$

$$\hat{Y}_{1t}(0) = \sum_{j=2}^{J+1} w_j Y_{jt}, \qquad \hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}(0)$$

  - $w_j \geq 0$: the weight of donor j, with $\sum_j w_j = 1$

The synthetic control estimator (2/2)

- **Weights**: the convex combination of donors closest to the treated unit in the pre-treatment predictors

$$w^*(V) = \arg\min_w (X_1 - X_0 w)' V (X_1 - X_0 w), \qquad w_j \geq 0, \ \sum_j w_j = 1$$

  - X_1: the vector of pre-treatment predictors of the treated unit (outcome averages, covariates); X_0: the matrix of the same predictors for the donors
  - a quadratic programme on the simplex: weights are sparse and interpretable; no extrapolation outside the donors' range

- $V = \mathrm{diag}(v)$: the importance of each predictor
  - chosen to minimise the pre-period $\mathrm{MSPE} = \frac{1}{T_0} \sum_{t \leq T_0} (Y_{1t} - \sum_j w_j^*(V) Y_{jt})^2$ (nested optimisation, Abadie, Diamond and Hainmueller (2011))
  - MSPE: mean squared prediction error of the outcome; alternative: cross-validation on a training and a validation period Abadie, Diamond and Hainmueller (2015)
  - with all pre-period outcomes in X, the covariates get no weight Kaul et al. (2022)

Why it works: the factor model

- Abadie, Diamond and Hainmueller (2010): the untreated outcomes follow a factor model (interactive fixed effects Bai (2009))

$$Y_{jt}(0) = \delta_t + \theta_t Z_j + \lambda_t \mu_j + \varepsilon_{jt}$$

  - δ_t: a common time effect; Z_j: observed covariates with time-varying coefficients θ_t; λ_t: unobserved common factors; μ_j: unobserved loadings of unit j; ε_{jt}: transitory shocks
  - DiD assumes λ_t constant (parallel trends); SC lets the common factors vary over time
- If the weights reproduce the treated unit before T_0, $\sum_j w_j Y_{jt} = Y_{1t}$ for all $t \leq T_0$ and $\sum_j w_j Z_j = Z_1$, then
  - the bias of $\hat{\tau}_{1t}$ is bounded by a term that **shrinks as T_0 grows**, relative to the scale of ε (Appendix)
  - a good fit over a long pre-period is evidence that the loadings μ_j are matched
- A short pre-period with a perfect fit can be overfitting the noise: the bound is then weak Abadie (2021)

Feasibility conditions

- **Convex hull** Abadie (2021)
  - the treated unit must lie inside the range of the donors; otherwise no convex combination fits it
- **Donor pool**
  - units with similar structure, not affected by the treatment (no spillovers), without large shocks of their own after T_0
- **No anticipation**
  - effects must not start before T_0; backdate T_0 to the announcement if needed
- **Long pre-period** with a good fit
  - **interpolation bias**: a mix of very different donors may not resemble the treated unit even if the averages match
- Report
  - weights, predictor balance, pre-period RMSPE, the placebo distribution, leave-one-out

Inference with one treated unit

- **In-space placebos** Abadie, Diamond and Hainmueller (2010): reassign the treatment to every donor in turn, refit, and compute the ratio

$$r_j = \mathrm{RMSPE}_{\mathrm{post}}/\mathrm{RMSPE}_{\mathrm{pre}}, \qquad \text{p-value} = \#\{j : r_j \geq r_1\}/(J+1)$$

  - RMSPE: root mean squared prediction error of the synthetic control, before (pre) and after (post) T_0; a large r_j: a large post-treatment gap relative to the pre-fit
  - $\#\{\cdot\}$: the number of units; the smallest attainable p-value is $1/(J+1)$; exact only under random assignment across units, otherwise a descriptive ranking Firpo and Possebom (2018)
- **Other robustness checks**
  - in-time placebos: a fictitious T_0 inside the pre-period should give no effect
  - leave-one-out: drop each donor with positive weight and refit
- **Conformal inference** Chernozhukov, Wüthrich and Zhu (2021): test H_0: $\tau_{1t} = \tau_0$ by permuting the residuals over time
  - block permutations for dependence; confidence sets by inverting the test over τ_0

Case study: the economic cost of German reunification



Brandenburg Gate, Berlin, 11 November 1989

Photo: Peer Grimm, Bundesarchiv Bild 183-1989-1111-009 (1989); CC BY-SA 3.0 de; Wikimedia Commons

- Abadie, Diamond and Hainmueller (2015)
  - did the 1990 reunification lower West German GDP per capita?
- Data Abadie, Diamond and Hainmueller (2014)
  - West Germany and 16 OECD countries, 1960–2003
  - predictors: GDP per capita, trade openness, inflation, industry share, schooling, investment rate
- V by cross-validation (Section 4 and the replication code)
  - training predictors 1971–1980, validation outcomes 1981–1990
  - final weights from the 1981–1990 predictors with that V
- Erratum (2026)
  - the outcome is GDP per capita in PPP *current* USD, not 2002 USD as labelled in the article

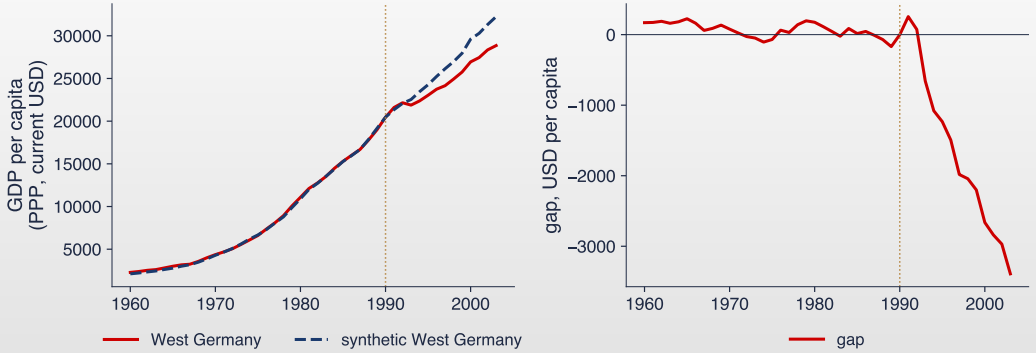
Replication: weights and predictor balance

Donor	ours	ADH, Table 1
Austria	0.42	0.42
United States	0.22	0.22
Japan	0.16	0.16
Switzerland	0.11	0.11
Netherlands	0.09	0.09

Predictor	West Germany	synthetic	OECD avg.
GDP per capita	15809	15805	13669
Trade openness	56.8	56.9	59.8
Inflation	2.6	3.5	7.6
Industry share	34.5	34.4	33.8
Schooling	55.5	55.3	38.7
Investment rate	27.0	27.0	25.9

- The cross-validated V and the quadratic programme reproduce the published weights to two decimals; all other donors get zero weight
- The simple OECD average differs on every predictor: the unweighted comparison would be biased

West Germany and its synthetic control



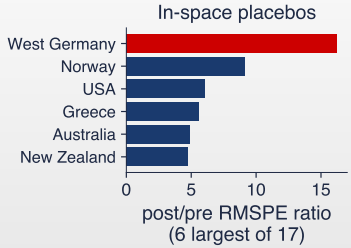
- Left: GDP per capita (PPP, current USD); right: the gap; pre-1990 RMSPE 121 USD

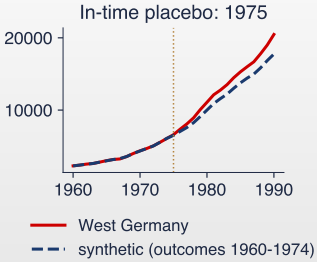
ATS_ch14_germany

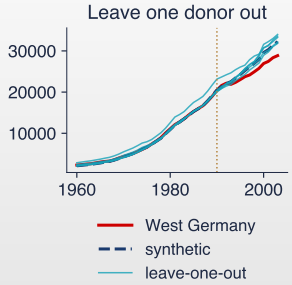
Interpreting the German replication

- The synthetic West Germany tracks the actual series for thirty years (1960–1989), then grows faster
- Average gap 1990–2003: -1588 USD per capita a year (-5.6% of the synthetic level); in 2003: -3391 USD (-10.5%)
- The paper reports a reduction of about 1,600 USD a year on average: the replication matches
- The donors (Austria, USA, Japan, Switzerland, Netherlands) are rich, industrial economies: the weights make economic sense, a check no algorithm does for us

Placebos for the German case







- Left: post/pre-1990 RMSPE ratio when each country is treated in turn (ADH 2015, Figure 6); centre: reunification moved to 1975 (as in Figure 4, outcome-only SC); right: one donor left out at a time (Figure 5)

ATS_ch14_germany

Interpreting the German placebos

- West Germany has the largest ratio (16.2; next: Norway, 9.2): permutation p-value $1/17 = 0.059$, the smallest attainable
- Placebo reunification in 1975 (outcome-only SC on 1960–1974, pre-RMSPE 19 USD): West Germany grows faster than its synthetic control (average gap +9.9% to 1990): no spurious cost appears
- With the published 1975 predictors the cross-validated V is not identified: one starting value gives Austria as the donor, the best of eight gives United States Klößner et al. (2018)
- Leaving out any donor keeps a negative gap in 2003, between 2965 and 5138 USD: the result does not hinge on one country
- With 17 units the permutation test cannot give $p < 0.059$

Case study: the Brexit doppelganger

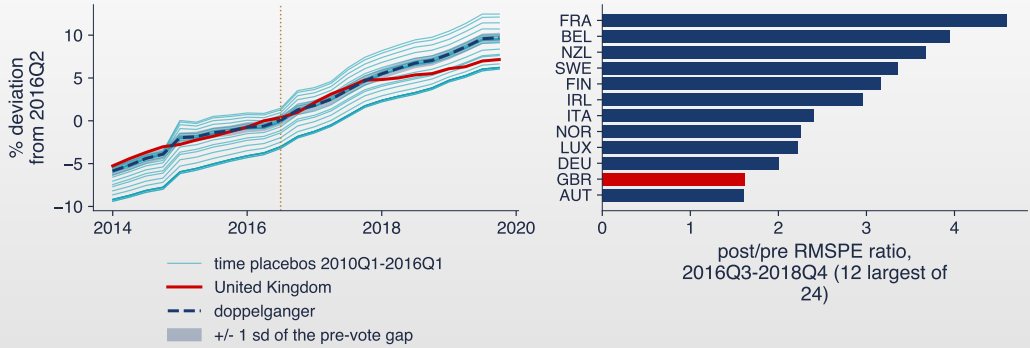


Counting the votes of the EU referendum, 23 June 2016

Photo: Steve Bowbrick (2016); CC BY 2.0; Wikimedia Commons

- Born et al. (2019)
  - the referendum of 23 June 2016 as a natural experiment; outcome: UK real GDP
- Design (Section 2.1)
  - 23 OECD donors; quarterly real GDP, normalised to 1 in 1995
  - pre-period 1995Q1–2016Q2; six covariate averages
  - published weights (Table 2): United States 0.51; Italy 0.17; New Zealand 0.14; Hungary 0.11; Germany 0.05
- Published result
  - UK output 2.4% below the doppelganger by the end of 2018
  - significant by the end-of-sample instability test of Andrews (2003)
- Our replication
  - the same donors and window; the current OECD vintage (to 2026Q2); the GDP path only (no covariates)

The doppelganger on today's data



- Left: UK real GDP and its doppelganger (% from 2016Q2), with the doppelgangers of fictitious votes in every quarter 2010Q1–2016Q1; right: the 12 largest post/pre RMSPE ratios of the 24 countries

ATS_ch14_brexit

Interpreting the Brexit replication

- Weights on today's data: United States 0.36; Japan 0.19; Canada 0.16; Portugal 0.10; Hungary 0.09; the doppelganger changes with the data vintage
- Gap: 0.1% in 2017Q4, -1.5% in 2018Q4 (published: -2.4%), -2.5% in 2019Q4; pre-vote standard deviation 0.51%
- The UK ranks 11 of 24 in the country placebos; the 25 time placebos give 2018Q4 gaps from -4.3% to 1.9%: on revised data the evidence is much weaker
- A replication changes the data (revisions), the specification (no covariates) and the inference (permutation instead of an end-of-sample test): report each deviation separately

Recap: Synthetic control

- A convex combination of donors fitted on a long pre-period; justified by a factor model; transparent sparse weights
- Inference by placebos in space and time, leave-one-out, conformal methods; $p \geq 1/(J+1)$
- German reunification replicates exactly; the Brexit gap is sensitive to data revisions and specification

Intercepts and augmentation

- **Demeaned SC** Doudchenko and Imbens (2016), Ferman and Pinto (2021): allows a constant level difference c between the treated unit and the donors

$$\hat{Y}_{1t}(0) = c + \sum_j w_j Y_{jt}$$

  - the weights are fitted on pre-period outcomes minus the pre-period mean of each unit
  - solves the level part of the convex-hull problem; consistent under imperfect pre-fit when the factors are stationary Ferman and Pinto (2021)

- **Augmented SC** Ben-Michael, Feller and Rothstein (2021): corrects the SC estimate by the remaining predictor imbalance, through a ridge regression

$$\hat{Y}_{1t}^{\text{aug}}(0) = \sum_j \hat{w}_j Y_{jt} + \Big(X_1 - \sum_j \hat{w}_j X_j\Big)' \hat{\eta}_t$$

  - $\hat{w}_j$: the SC weights; X_j: the pre-period outcomes of unit j; $\hat{\eta}_t$: coefficients of a ridge regression of Y_{jt} on X_j across donors, with penalty λ
  - equivalent to weights $\hat{w} + X_0(X_0'X_0 + \lambda I)^{-1}(X_1 - X_0'\hat{w})$, which can be negative (controlled extrapolation)
  - $\lambda \to \infty$ gives SC; $\lambda \to 0$ gives an outcome regression; the bias falls with the remaining imbalance

Synthetic difference in differences (1/2)

- Arkhangelsky et al. (2021): a two-way fixed-effects regression in which units and periods are weighted

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min \sum_{j,t} \left(Y_{jt} - \mu - \alpha_j - \beta_t - W_{jt}\tau\right)^2 \hat{\omega}_j \hat{\lambda}_t$$

  - $W_{jt} = 1$ for the treated unit after T_0, 0 otherwise; τ: the effect; μ: the overall mean; α_j: unit fixed effects; β_t: period fixed effects
- **Unit weights** $\hat{\omega}_j$: SC with an intercept and a ridge penalty
  - penalty $\zeta^2 T_0 \|\omega\|^2$ with $\zeta = (N_{\text{tr}} T_{\text{post}})^{1/4} \hat{\sigma}$; N_{tr}: treated units; T_{post}: post-periods; $\hat{\sigma}$: the noise level of the controls
- **Time weights** $\hat{\lambda}_t$: the pre-periods that best predict the post-period mean of the controls

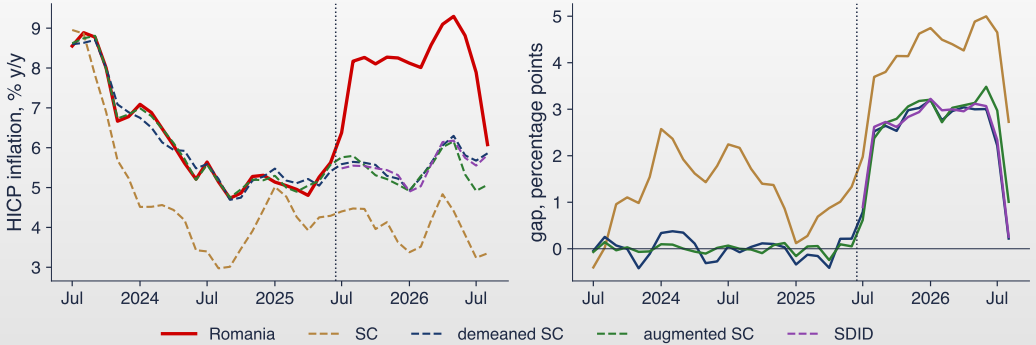
Synthetic difference in differences (2/2)

- Three estimators as special cases
  - DiD: ω and λ uniform
  - SC: unit weights ω only (uniform λ), no unit fixed effect
  - SDID: both sets of weights, both fixed effects
- One treated unit: placebo standard error
  - each control in turn plays the treated unit; the standard deviation of the placebo estimates is the standard error (Algorithm 4)

Romania 2025: design of the synthetic control

- Outcome: annual HICP inflation (pp), $100(\ln P_t - \ln P_{t-12})$, Eurostat; treated: Romania; donors: the other 26 EU countries
  - P_t: the HICP price index in month t; pp: percentage points
  - the annual rate removes seasonality and makes a price-level shock visible for exactly twelve months
- Pre-period: July 2023 – June 2025 (24 months, after the 2021–2023 surge); treatment from July 2025; effect window July 2025 – June 2026
  - a built-in falsification: once the price-level jump leaves the 12-month window (August 2026), the gap must close
- Estimators: SC, demeaned SC, ridge-augmented demeaned SC, SDID; inference: in-space placebos
- Threats: other countries' own energy and tax measures; anticipation of the VAT increase; Romania outside the donors' range

Four counterfactuals for Romanian inflation



⊡ Left: Romania and the four estimated counterfactuals; right: the gaps; vertical line: June 2025; data to August 2026

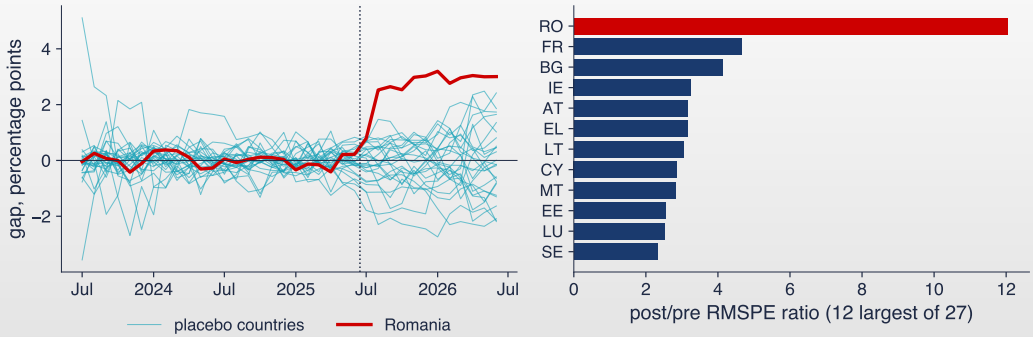
ATS_ch14_romania

Interpreting the four estimators

Estimator	pre RMSPE	August 2025	avg. Jul 2025 – Jun 2026	August 2026
SC	1.45	3.70	4.18	2.73
demeaned SC	0.23	2.53	2.70	0.22
augmented SC	0.09	2.38	2.78	1.01
SDID	–	2.62	2.75 (*se* 0.70)	0.26

- Classic SC puts all weight on Croatia 0.85; Hungary 0.15: Romania is outside the convex hull, the pre-fit is poor and the effect is overstated
- With an intercept the fit is good (weights France 0.38; Slovenia 0.25; Sweden 0.13; Malta 0.10, ...); augmented SC ($\lambda = 1$, 9 negative weights) and SDID agree: 2.7–2.8 pp over twelve months
- Falsification: in August 2026 the gap falls to 0.22 (demeaned SC) and 0.26 (SDID); the augmented SC keeps 1.01, the price of extrapolating with negative weights

Placebo inference for Romania



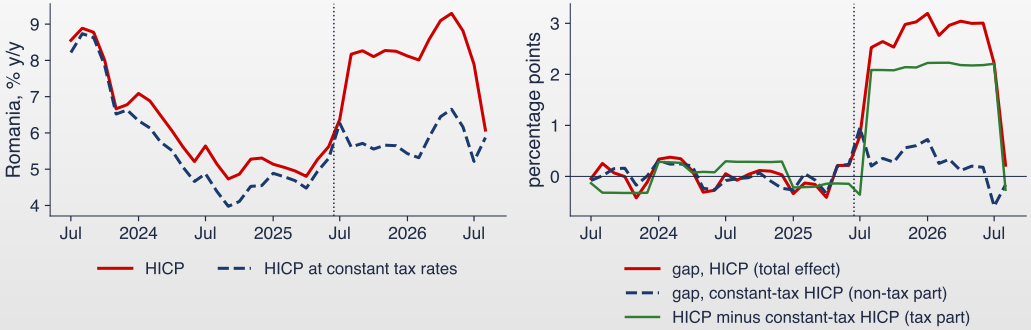
- Demeaned SC applied to every EU country in turn; left: gaps; right: the 12 largest post/pre RMSPE ratios over July 2025 – June 2026

ATS_ch14_romania

Interpreting the Romanian placebos

- Romania has the largest ratio (12.0; next France, 4.7): $p = 1/27 = 0.037$
- No placebo country shows a jump of this size in July–August 2025: the effect is not a common European shock
- The permutation p-value treats Romania as one of 27 exchangeable units; it is a measure of rarity, not a sampling-based confidence statement
- Other countries' measures in the window (tax changes, the end of their own caps) bias the effect towards zero if they raised inflation there

The tax component of Romanian inflation



⊡ Left: Romanian HICP and HICP at constant tax rates (Eurostat (2026b)); right: the total gap, the gap of the constant-tax inflation (same demeaned SC on all countries) and the tax wedge

ATS_ch14_romania

Interpreting the tax decomposition

- The constant-tax index assumes full and immediate pass-through of tax changes: the tax wedge jumps by 2.09 pp in August 2025, when the VAT rose
- Average over twelve months: total 2.70 pp = tax wedge 1.95 pp + constant-tax gap 0.40 pp: about 72% of the effect is mechanical tax
- The non-tax part peaks in July 2025 (0.96 pp): the end of the electricity cap, which is a price, not a tax
- Pass-through of VAT to consumer prices is often incomplete and asymmetric Benedek et al. (2020); the constant-tax index is an accounting benchmark, not a measured pass-through

Staggered adoption and two-way fixed effects (1/2)

- Many units adopt at different dates g (their cohort); the static TWFE regression

$$Y_{it} = \alpha_i + \beta_t + \tau D_{it} + u_{it}$$

  - $D_{it} = 1$ if unit i is treated at t; α_i, β_t: unit and period fixed effects; τ: a single treatment effect
- Goodman-Bacon (2021): $\hat{\tau}$ is a weighted average of all 2×2 DiDs
  - including "forbidden" ones that use already-treated units as controls
  - with effects that grow over time some weights are negative de Chaisemartin and D'Haultfœuille (2020): $\hat{\tau}$ can even have the wrong sign

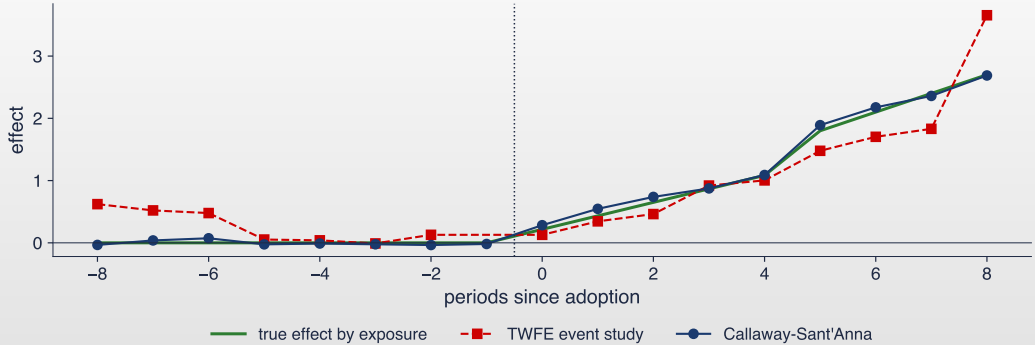
Staggered adoption and two-way fixed effects (2/2)

- Callaway and Sant'Anna (2021): group-time effects, each compared only with never-treated units

$$ATT(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid \text{never treated}]$$

  - G: the adoption date of a unit; Y_{g-1}: the outcome in the last period before adoption; ATT: average treatment effect on the treated
  - then aggregate by exposure $(t-g)$ or by cohort
- Related: interaction-weighted event studies Sun and Abraham (2021); a guide to the new DiD literature Roth et al. (2023)
- Pre-trend tests have low power; report the sensitivity of the effect to violations of parallel trends

TWFE against Callaway and Sant'Anna



- Simulated panel: three cohorts adopting in periods 6, 11 and 16 and a never-treated group; effects grow with exposure, faster for early adopters

ATS_ch14_did_dml

Interpreting staggered adoption

- Static TWFE: 0.55; true average effect on the treated: 1.99; Callaway–Sant'Anna: 2.04
- TWFE uses early adopters as controls for late adopters while their effect is still growing: those comparisons subtract part of the effect
- The TWFE event study is distorted at long exposures (binned endpoint 3.66 against 2.70); CS gives 2.69
- With one treated country (Romania) there is no staggering; with EU-wide policies adopted at different dates, there is

Recap: Beyond synthetic control

- An intercept fixes level differences; ridge augmentation corrects the remaining imbalance; SDID combines unit and time weights
- Romania 2025: 2.7–2.8 pp more annual inflation for twelve months, most of it the mechanical VAT effect, and a falsification that holds
- Staggered adoption breaks TWFE; use group-time estimators

The CausalImpact model (1/2)

- Brodersen et al. (2015): the outcome is a trend plus a seasonal component plus a regression on control series

$$y_t = \mu_t + \gamma_t + \beta' x_t + \varepsilon_t, \qquad \mu_{t+1} = \mu_t + \delta_t + \eta_t, \qquad \delta_{t+1} = \delta_t + \zeta_t$$

  - μ_t: local level; δ_t: local slope; γ_t: seasonal component; x_t: control series with coefficients β; $\varepsilon_t, \eta_t, \zeta_t$: independent Normal noises
- A structural time series model Harvey (1989) in state space form, filtered and smoothed by Kalman (Chapter 6, Durbin and Koopman (2012))
  - a spike-and-slab prior on β selects the controls Scott and Varian (2014); the posterior is computed by MCMC

The CausalImpact model (2/2)

- Fit on the pre-period; simulate the post-period counterfactual $y_t^{(s)}(0)$ from the posterior predictive, given the observed x_t
  - s: the index of a simulated path; effects $y_t - y_t^{(s)}(0)$, pointwise and cumulative
  - credible intervals from the quantiles of the simulated effects
- **Identification**
  - the controls are **not affected** by the treatment
  - their relation with y is stable after T_0

Our implementation

- `statsmodels` `UnobservedComponents`: local level + regression on the controls, maximum likelihood on the pre-period
  - 1000 counterfactual paths: parameters drawn from their asymptotic normal distribution, then the state simulated forward from the end of the pre-period given the observed controls
- Differences from the R package: no spike-and-slab selection, no MCMC; parameter uncertainty by a Gaussian approximation
  - the transparent version keeps every step visible; with few controls the two give similar intervals
- Outputs: average effect with a 95% interval, cumulative effect, and the tail probability that the counterfactual mean exceeds the observed one

Case study: the US spot Bitcoin ETFs

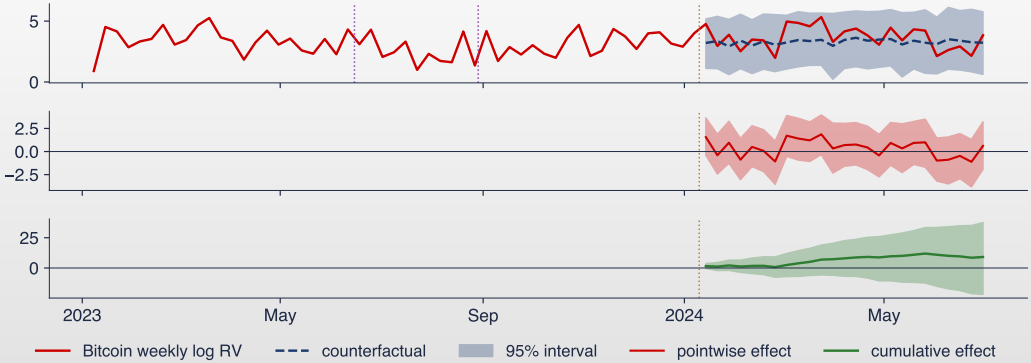


A Bitcoin ATM, Prague

Photo: Perituss (2016); CC0; Wikimedia Commons

- **10 January 2024** SEC (2024)
  - the SEC approves eleven spot Bitcoin ETPs (exchange-traded products); trading starts on 11 January
- Question
  - did the approval change Bitcoin volatility (institutional demand, arbitrage between spot and ETF)?
- Design
  - outcome: weekly log realised variance of Bitcoin
  - controls: the same measure for the S&P 500, Nasdaq 100, gold and EUR/USD
  - pre-period January 2023 – 7 January 2024 (53 weeks); post: 25 weeks to June 2024
- Anticipation (dotted lines)
  - BlackRock filed on 15 June 2023; a court ruled for Grayscale on 29 August 2023

CausalImpact for the spot ETF approval



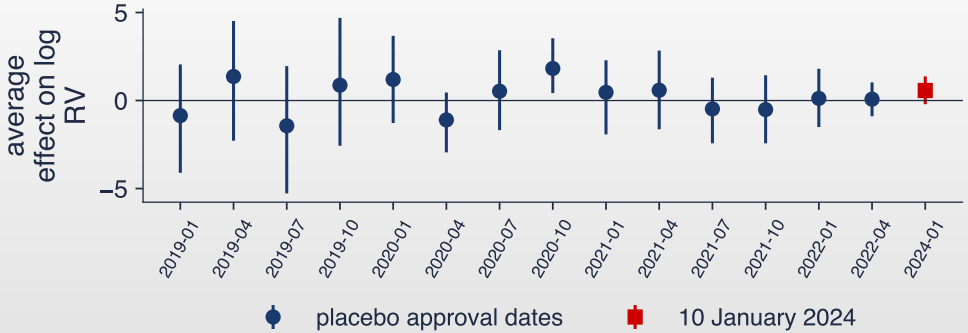
Top: observed and counterfactual log RV with a 95% interval; middle: pointwise effect; bottom: cumulative effect

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Interpreting the ETF analysis

- Average effect on weekly log RV: 0.36 (95% interval [−0.87, 1.50]), tail probability 0.26: no detectable change
- The level variance is small (0.011) against the irregular one (0.94): weekly crypto volatility is mostly noise, and the controls explain little
- The finding agrees with a peer-reviewed study: no effect on Bitcoin's own volatility Babalos, Bouri and Gupta (2025)
- Absence of evidence is not evidence of absence: with this noise, a 50% change in volatility would not be detected

Placebo dates and the choice of controls



- The same analysis with 14 fictitious approval dates in 2019–2022 (53 pre-weeks, 25 post-weeks each) and the true date with the same window lengths

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Interpreting the placebo dates

- Only 1 of the 14 placebo intervals excludes zero; the placebo estimates have a standard deviation of 0.97: the real estimate is inside their range
- With Ether as an extra control: 0.18 [−0.47, 0.87]; but Ether may itself respond to the approval: a treated control biases the effect towards zero
- Moving T_0 back to the BlackRock filing (June 2023): −0.02 [−1.74, 1.62]: still nothing, so anticipation does not hide a large effect
- Placebo dates calibrate the credible intervals: if many placebos look "significant", the model is misspecified

Recap: BSTS and CausalImpact

- A state space forecast of the untreated path from control series; uncertainty from parameters and states
- Its key assumption is about the controls: unaffected and stably related; check with placebo dates
- The spot ETF approval did not change Bitcoin volatility detectably; the design could not see small effects

Double/debiased machine learning for time series (1/2)

- Chernozhukov et al. (2018): the **partially linear model**

$$y_t = \theta d_t + g(X_t) + u_t, \qquad d_t = m(X_t) + v_t$$

  - d_t: the treatment; θ: its effect; X_t: control variables; g, m: unknown **nuisance functions**, learned by machine learning; u_t, v_t: errors

- **Orthogonal score**: regress the residual of y on the residual of d

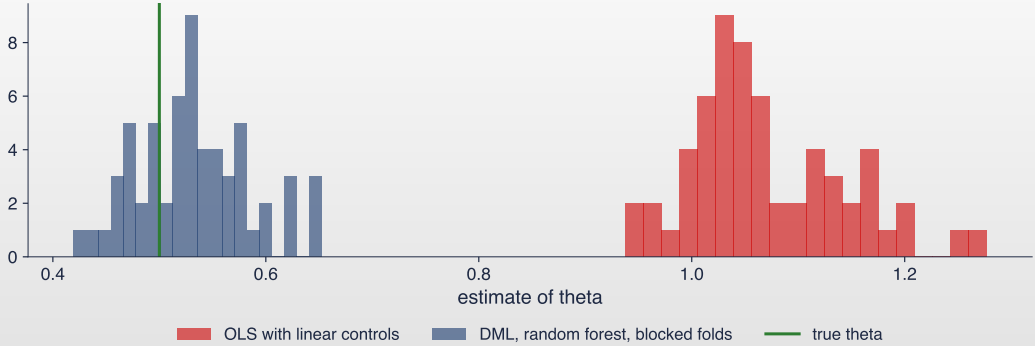
$$\psi = \big(y_t - \ell(X_t) - \theta(d_t - m(X_t))\big)\big(d_t - m(X_t)\big), \qquad \ell(X) = \mathbb{E}[y \mid X]$$

  - $\hat{\theta}$ solves $\frac{1}{T}\sum_t \psi_t = 0$; first-order insensitive to errors in ℓ and m

Double/debiased machine learning for time series (2/2)

- **Cross-fitting**: the nuisance functions are fitted on other folds than the one where ψ is evaluated
  - for time series use **contiguous blocks** with a gap around the held-out block
  - standard error from a HAC variance of ψ
- DML removes regularisation and overfitting bias; it does not create identification
  - d_t must be unconfounded given X_t

DML against a linear adjustment



- Simulated dependent data: 10 VAR(1) covariates, nonlinear g and m, AR(1) errors, $\theta = 0.5$, $T = 1\,000$, 60 replications; random forests with five blocked folds

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Interpreting the DML experiment

- OLS with linear controls: mean estimate 1.07, far from 0.5: the nonlinear confounding is not removed
- DML: mean 0.53, standard deviation 0.05; 95% HAC intervals cover the true value in 88% of the replications
- The remaining bias comes from the forests' errors on a finite sample; it shrinks with T
- In macro time series the treatment is rarely unconfounded given observables: DML is a tool for the estimation step, after identification is argued

Identification, method by method

Method	Key assumption	Check
Granger, PCMCI	no hidden common causes; correct timing; faithfulness	add candidate confounders; change the frequency
ITS, event study	nothing else changes at T_0; no anticipation	placebo dates; news timeline
Synthetic control, SDID	factor model; donors unaffected; treated unit in the donors' range	pre-fit, placebos, leave-one-out
DiD (staggered)	parallel trends; no anticipation	pre-trends with sensitivity analysis
CausalImpact	controls unaffected; stable relation	placebo dates; alternative control sets
SVAR, LP, DML	as good as random given the past or an instrument	narrative evidence; instrument strength

Transparent reporting

- State the estimand (which effect, on which unit, over which horizon) before choosing the method
- Fix treatment dates, windows, donors and estimators before looking at post-treatment data; publish the pre-registration
- Report every placebo, every specification and every deviation from the published design; a specification curve shows them together
- Separate statistical uncertainty (intervals) from identification uncertainty (assumptions); the second is usually larger
- Read Athey and Imbens (2017) and Abadie (2021) before writing a policy evaluation

An open question

- The question
  - how much of Romania's 2025–2026 inflation surge was caused by the end of the electricity cap and the VAT increase?
  - how robust is the answer to the choice of method?
- A testable form
  - formal: H_0: the average gap over July 2025 – June 2026 is zero; the claim "between 2 and 3.5 pp" must hold for every pre-registered estimator with a good pre-fit
  - falsified if a reasonable specification gives less than 2 pp or a placebo country matches Romania
- Why it matters
  - monetary policy must tell one-off price-level effects from persistent inflation
  - second-round effects decide interest rates
  - literature to start from: Abadie, Diamond and Hainmueller (2015), Ben-Michael, Feller and Rothstein (2021), Arkhangelsky et al. (2021), Benedek et al. (2020)

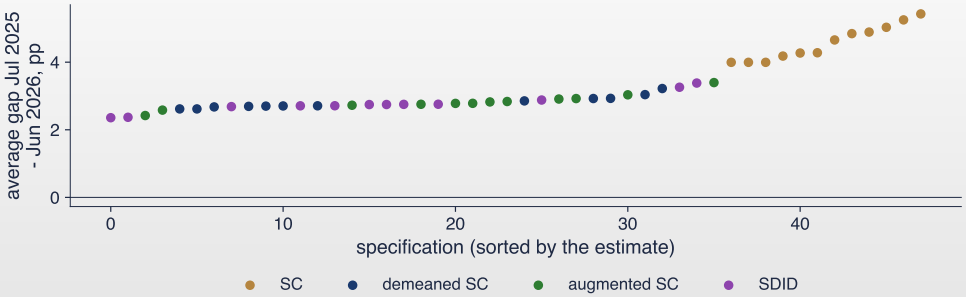
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed studies since 2015 that estimate VAT pass-through to consumer prices with synthetic control or DiD; give DOIs and the estimated pass-through.` Then check every DOI on Crossref
  - **design**: `Write a pre-registration: outcome, donors, exclusions, pre-period, estimators, placebo tests, what counts as a robust effect.`
  - **code and replication**: reproduce first the German weights (Austria 0.42; USA 0.22; ...) before running anything on Romania
  - **critique**: `Act as a hostile referee: which EU countries had their own energy or tax measures in 2025-2026, and how would each bias the estimate?`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- Policy dates are taken from official sources (Monitorul Oficial, Eurostat, SEC), not from the assistant's memory
- The donor pool excludes countries with their own large shocks in the window, and the exclusions are written before the results are seen
- Placebo tests are run with exactly the same code and settings as the main estimate
- An AI claim that "the synthetic control proves the VAT caused 3 pp of inflation" is rewritten as an estimate with assumptions

Mini-case: a specification curve for Romania



- 48 specifications: four estimators × three donor pools (EU-26, euro area, non-euro EU) × four pre-periods (from 2017, 2019, July 2023, 2024; the 2021–2023 surge excluded)
- Without classic SC: from 2.36 to 3.40 pp, median 2.75; classic SC: median 4.47 (poor pre-fit); 0 specifications give a non-positive effect: the hypothesis survives

ATS_ch14_ai_case

Project idea

- **Tax pass-through in Central and Eastern Europe with synthetic controls**: replicate first, then extend
  - replicate: Abadie, Diamond and Hainmueller (2015) (Table 1, Figures 2–5) with the public data, then this chapter's Romanian estimate
  - extend: product-level HICP (food, energy, services) to measure pass-through by VAT rate
  - other episodes (Hungary 2012, Croatia 2023 euro adoption); conformal intervals Chernozhukov, Wüthrich and Zhu (2021)
  - pre-register: donors and exclusions, windows, estimators, placebos, the robustness rule
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Granger causality and its relatives (TE, PCMCI, CCM) describe predictive structure; causal claims need assumptions about the assignment
- Interrupted series and event studies use the unit's own past; dependence enlarges their standard errors
- Synthetic control and its successors use other units; placebos are the inference, a long pre-fit is the justification
- Replication exposes fragility: German reunification holds, the Brexit gap depends on data vintage
- Romania 2025: a robust 2.7–2.8 pp twelve-month effect, mostly mechanical VAT; Bitcoin ETFs: no detectable effect on volatility

Self-assessment

Questions

- Why can adding a variable to the information set remove a Granger causality?
- What does conditioning on the parents of the source achieve in the MCI test?
- Why does classic synthetic control fail for Romanian inflation?
- What is the smallest placebo p-value with 23 donors?
- Why can a static TWFE estimate have the wrong sign under staggered adoption?

Next:

- Review and project defence
- one synthesis per chapter; the replication project and the oral defence

Appendix: the bias bound of synthetic control

- Factor model $Y_{jt}(0) = \delta_t + \lambda_t \mu_j + \varepsilon_{jt}$ (F factors); weights with $\sum_j w_j Y_{jt} = Y_{1t}$, $t \le T_0$
- Stack the pre-period: $\lambda^P(\mu_1 - \sum_j w_j \mu_j) = -(\varepsilon_1^P - \sum_j w_j \varepsilon_j^P)$; if $\lambda^{P\prime}\lambda^P$ is invertible, solve for the loading imbalance
- Then $Y_{1t}(0) - \sum_j w_j Y_{jt} = \lambda_t (\lambda^{P\prime}\lambda^P)^{-1}\lambda^{P\prime} \sum_j w_j \varepsilon_j^P - \cdots + (\varepsilon_{1t} - \sum_j w_j \varepsilon_{jt})$
- The first term has mean zero only approximately (the weights depend on the ε^P); ADH (2010, Appendix B) bound it by a quantity of order $J^{1/p} \bar{m}_p^{1/p} F \bar{\lambda}^2 / (\xi T_0^{1-1/p})$: small when T_0 is large relative to J

Back

Appendix: Gaussian transfer entropy equals half the Geweke measure

- For jointly Gaussian variables $I(Y;X \mid Z) = \frac{1}{2} \ln \dfrac{\mathrm{Var}(Y \mid Z)}{\mathrm{Var}(Y \mid X, Z)}$ (the entropy of a Gaussian is $\frac{1}{2}\ln(2\pi e\sigma^2)$)
- With $Y = y_t$, $X = x_{t-1}^{(l)}$, $Z = y_{t-1}^{(k)}$, the conditional variances are the residual variances of the restricted and full Granger regressions
- Hence $TE_{x\to y} = \frac{1}{2} F_{x\to y}$, and $2T \cdot \widehat{TE} \approx T\hat{F} \approx W$, the likelihood-ratio form of the Granger Wald test Barnett, Barrett and Seth (2009)
- Non-Gaussian data break the equality: TE then detects dependence that a linear test misses (Section 1)

Back

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